

Part 1: (100 points)

I. International Competition Type Questions (25 points in total)

Note: For questions with five options (questions 1 and 2), 0 points are awarded for all incorrect answers, and 0.5 , 1, 2, 4, and 7 points are awarded for each correct answer (options 1, 2, 3, 4, and 5

respectively). For questions with four options (questions 3 and 4), 0 points are awarded for all incorrect answers, and 0 points are awarded for each correct answer (options 1, 2, 4, and 7 respectively).

The second, third, and fourth items are worth **0.5, 1, 2.5** , and 5.5 points respectively.

Question 1:

In November 2019, the Centers for Disease Control (CDC) announced the first confirmed case of Severe Fever with Thrombocytopenia Syndrome (SFTS) in Taiwan. The patient had sought medical attention at a clinic due to persistent fever and vomiting, and was later hospitalized due to rash and altered mental status. The case was initially reported as a suspected case of dengue fever, but subsequent tests confirmed a negative dengue fever result. However, the patient tested positive for SFTS virus (SFTSV). SFTSV belongs to the Bunyaviridae family.

The tick-borne virus (Bunyaviridae) first appeared in mainland China in 2009, and was first reported in 2011. Subsequent outbreaks have occurred...

In Japan and South Korea, the mortality rate is 5-15%, and the main route of transmission is via Haemaphysalis ticks.

Bites from ticks carrying SFTSV (shorticorns). The incubation period is approximately 7-14 days. After a human is bitten by a tick carrying SFTSV, the bite site...

Symptoms may include redness, swelling, rash, blisters, or ecchymosis. Patients may also experience fever, nausea, vomiting, loss of appetite, and decreased platelet and white blood cell counts. A small number of patients may experience multiple organ failure and even death. There is currently no specific treatment for SFTS.

However, supportive care can reduce the mortality rate. Prevention includes avoiding lying down or sitting for extended periods in grassy areas, wooded areas, etc.

When engaging in activities in these areas, wear light-colored long-sleeved clothing and trousers to protect against tick bites. The SFTSV gene has three segments: large (L), medium (M), and small (S), which encode RNA-dependent RNA polymerase (RdRp), glycoprotein precursor (M), glycoprotein N (Gn), and sugar...

There are six types of proteins, including protein C (Gc), nucleoprotein (NP), and non-structural protein (NS).

Please determine whether the following five statements are correct, and fill in T (for correct) or F (for incorrect) in the corresponding column of the table below.

(A) In addition to SFTS, ticks are also vectors for diseases such as Q fever, scrub typhus, tick-borne encephalitis, and Lyme disease.

(B) In addition to tick bites, contact with the blood, urine, cerebrospinal fluid, or tracheal aspiration fluid of patients in the acute phase can also lead to infection.

SFTS infection.

(C) Ticks mostly live in the wild and are the source of emerging infectious diseases in humans, such as SFTSV.

The head belongs to some kind of wild animal.

(D) There have never been any reports of Haemaphysalis longicornis ticks in Taiwan, so apart from the large number of imported cases following infection from overseas, it is unlikely that...

There has been an outbreak of SFTS.

(E) The lack of nucleic acid proofreading function in the RdRp of SFTSV leads to increased gene mutations. Furthermore, gene segmentation increases the probability of genomic

fragment exchange among viral members. Gene rearrangement increases the virus's adaptability to the environment and its pathogenicity.

Its strength and increased host range pose a threat to human public health.

Narrative	(A)	(B)	(C)	(D)	(AND)
T or F					

Question 2: The

2019 Nobel Prize in Physiology or Medicine was awarded to William G. Kaelin Jr., Sir Peter J. Ratcliffe, and Gregg L. Semenza for their work on how cells sense and adapt to changes in oxygen concentration. They discovered that cells...

Animals can utilize molecular mechanisms to regulate gene activity to adapt to varying oxygen concentrations. Animals employ different mechanisms to regulate tissue and...

Cells obtain oxygen. Otto Warburg discovered that enzymes in the mitochondria are needed to utilize oxygen in order to convert food into energy.

Corneille Heymans was awarded the Nobel Prize in 1931 for his discovery of the carotid body in the human carotid artery.

The carotid body contains specific cells that can sense the oxygen content in the blood, thereby influencing the regulation of the brain's respiratory center.

Respiratory rate, for which he received the 1938 Nobel Prize. Semenza first identified the hypoxia-inducible factor (HIF) regulating oxygen-dependent responses. In oxygen deficiency, it can activate certain genes. After purification, he discovered that HIF consists of two parts: the oxygen-sensitive HIF-1 α and the non-oxygen-regulated protein ARNT. Kaelin discovered...

Patients with von Hippel-Lindau syndrome (VHL) are prone to developing certain cancers, as seen in isolated...

Following the introduction of the VHL tumor suppressor gene, it was found to inhibit the growth of VHL cell line tumors; Ratcliffe discovered that VHL regulates...

Post-translational oxygen-sensitive degradation of HIF-1 α : Under low oxygen conditions, HIF-1 α is protected from degradation and accumulates in the nucleus. After binding to ARNT, it can bind to specific DNA sequences (HREs) in hypoxia-regulating genes. Under normal oxygen conditions, HIF-1 α is degraded by proteases, with oxygen regulating degradation through the addition of hydroxyl groups (OH) to HIF-1 α . VHL

The protein can recognize and form a complex with HIF-1 α , leading to degradation. Oxygen sensing allows cells to adjust their biological processes when oxygen levels are low.

The physiological mechanisms, or the generation of new blood vessels and new red blood cells, also control angiogenesis and placental development during fetal development.

Stimulating angiogenesis in tumors is a mechanism for cancer cell proliferation; drugs can be developed by stimulating or blocking oxygen sensing mechanisms.

These substances offer hope for combating diseases such as anemia and cancer in the future.

Please determine whether the following five statements are correct, and fill in T (for correct) or F (for incorrect) in the corresponding column of the table below.

(A) The carotid body contains chief cells and supporting cells. The chief cells can detect the oxygen content in the blood and secrete nerve endings.

Conductive substances regulate the frequency of breathing.

(B) When oxygen levels are low, the lungs secrete erythropoietin (EPO), which can act on the blood in the bone marrow.

In stem cells, it promotes the production of red blood cells.

(C) Animal cells can respond to changes in oxygen concentration at the gene level, through the expression of HIF transcription factors, allowing...

Cells can respond directly and rapidly to changes in oxygen pressure.

(D) The sensor that detects oxygen concentration in cells is HIF-1 α , which is degraded when oxygen concentration is normal and enters the cell nucleus when oxygen concentration is low, where it binds to a special transcription factor HRE.

(E) Hydroxylated HIF-1 α undergoes a structural change and becomes bound by VHL. The bound HIF-1 α can be pantothenic.

It is degraded by a protein degradation mechanism labeled with ubiquitin .

Narrative	(A)	(B)	(C)	(D)	(AND)
T or F					

Question 3:

In the past, some scientists believed that the lymphatic system originated from specialized stem cells of angioblasts.

Some scientists believe that the lymphatic system develops from cells derived from the main vein in early embryonic development and then migrates from there.

They then migrate to other tissues and organs within the embryo. Recent research from the Spanish Cardiovascular Research Center shows that the development of cardiac lymphatic vessels is not of a single origin, but rather involves the collaborative formation of cells from different tissues. This is essential for maintaining heart function.

In addition to the coronary arteriovenous system, the missing components should also include the lymphatic vessels of the heart. The function of the cardiac lymphatic vessels includes protecting...

Protecting the heart from infection, preventing myocardial edema, and aiding in the recovery of cardiac function after infarction are all important functions. They studied the origin of the cardiac lymphatic system by examining the formation of the mouse embryonic heart, showing that another type of lymphocyte emerges later in heart development.

The group was mobilized, originating from the second heart field (SWF) rather than from the main vein.

The two cardiac regions are composed of a multifunctional group of cells capable of producing different types of cardiac cells, including cardiomyocytes and smooth muscle cells. Muscle cells, endothelial cells of arteries and veins, and also lymphatic endothelial cells. Lymph produced in the second cardiac region.

Cells mix with lymphocytes from different sources, and these two groups of cells together form the lymphatic vessels on the ventral side of the heart. Second...

The cardiac region contributes 50% of lymphatic cells, but if their formation is blocked, the formation of cardiac lymphatic vessels will be completely prevented.

The results showed that this newly discovered cell population not only contributes to most of the cells in the cardiac lymphatic system, but also plays a significant role in the cardiac lymphatic system.

It plays an irreplaceable role in the formation of the system.

Please determine whether the following four statements are correct, and fill in T (for correct) or F (for incorrect) in the corresponding column of the table below.

(A) Since the cells in the second cardiac region can produce endothelial cells for arteries, veins, and lymphatic vessels, the second cardiac region should be...

It comes from the endoderm of the embryo.

(B) The development of general organs involves coordination between different types of cells, including the chemoradiography caused by perivascular cells.

The microenvironment is extremely important for its formation.

(C) In different cell subpopulations, regulatory errors can affect cell differentiation, specialization, and migration.

This can lead to defects in morphological development.

(D) The cardiovascular system develops from the precursor cells of the heart and is the earliest organ system to form during embryonic development.

The visceral lymphatic vessels develop from angiogenic cells originating from the second cardiac region.

Narrative	(A)	(B)	(C)	(D)
T or F				

Question 4:

Angiotensin-converting enzyme (ACE) can convert angiotensin.

(Angiotensin, Ang) I is an active form of Ang II, which causes vasoconstriction and increases blood pressure. Once the effect is complete, ACE2 is needed to remove one amino acid from the C-terminus of Ang I and Ang II, rendering them ineffective. ACE2 was first discovered in the heart and kidneys, and later found in the lungs, mainly distributed in type II alveolar cells, with a small amount distributed elsewhere.

In type I alveolar cells, tracheal epithelial cells, fibroblasts, endothelial cells, and macrophages. For more than four months, the COVID-19 pandemic has been fierce, surpassing the severe acute respiratory syndrome (SARS) caused by SARS-CoV in 2003. Main symptoms include fever, cough, and shortness of breath. The International Committee on Virus Nomenclature...

The committee named the pathogen SARS-CoV-2. Both SARS-CoV and SARS-CoV-2 are generally distinguished by their outer membrane.

The spike protein, protruding from the upper surface of the cell, binds to receptor proteins on the host cell surface and undergoes endocytosis. Subsequently, the coronavirus's mantle fuses with the outer membrane of its vesicles, releasing its RNA into the host cell for proliferation.

The spike protein of SARS-CoV-2 also infects cells via ACE2 on the cell surface. ACE2 is originally a transmembrane protease, which initiates a renin-angiotensin-vasoconstriction reaction involving ACE and ACE2 in cases of massive blood loss or a drop in blood pressure.

The renin-angiotensin system (RAS), ACE2 is normally located on the cell surface, but in SARS-CoV...

After entering and proliferating through surface ACE2, SARS-CoV develops numerous spike proteins within the cell.

Spiny proteins bind to free ACE2 within cells, thereby reducing the amount of ACE2 expressed on the cell surface. When cells...

When the amount of ACE2 on the cell surface decreases, Ang II cannot be metabolized properly, and the accumulation of Ang II can cause pulmonary vascular...

Increased permeability allows inflammatory substances to flow into the alveoli, causing acute pneumonia.

Determine whether the following four statements are true or false, and fill in T (true) or F (false) in the corresponding column of the table below.

(A) SARS-CoV invades cells using ACE2 as a receptor, so the direction of new drug development should be to inhibit ACE2 synthesis or

Drugs that inhibit the delivery of ACE2 to the cell membrane can help prevent SARS-CoV infection.

(B) Immunity acquired from SARS infection can effectively resist COVID-19 pneumonia; it can interact with SARS-CoV-2.

Protein-bound antibodies can effectively block the binding of SARS-CoV-2 to ACE2.

(C) The main symptoms of Middle East Respiratory Syndrome (MERS), which was prevalent in 2012, also included fever, cough, and shortness of breath.

The pathogen is also a coronavirus, which also has an outer membrane and single-stranded positive-sense RNA.

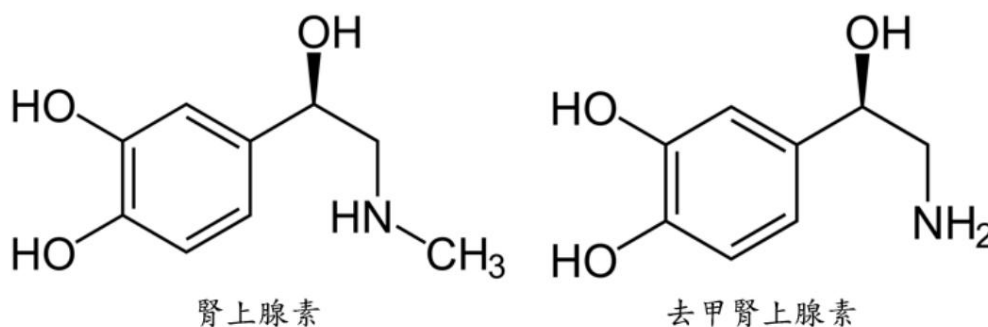
(D) A decrease in ACE2 on alveolar cells leads to an increase in the concentration of Ang II, which increases pulmonary vascular permeability.

Inflammatory substances can flow into lung tissue, triggering an excessive inflammatory response and causing organ damage.

Narrative	(A)	(B)	(C)	(D)
T or F				

II. True or False Questions (2.5 points **per blank** , **10 points in total**)

Catecholamines are a collective term for amine compounds containing catecholamines, which are produced by the adrenal medulla. (Adrenal medulla) produces a class of stress-responsive sympathomimetic "fight-or-flight" hormones. In the human body... Adrenergic receptors are a class of G protein-paired receptors that are stimulated by catecholamines. The G protein-coupled receptor (GPCR) receives catecholamines primarily adrenaline. Epinephrine (Ad) and norepinephrine (NAd). Adrenergic receptors are broadly divided into... γ_1 , γ_2 , γ_1 , γ_2 , and γ_3 . γ_1 receptors are mainly located on vascular smooth muscle, γ_2 receptors are located presynaptic, and γ_1 ... γ_2 receptors are located in the heart, γ_2 receptors in bronchial smooth muscle, and γ_3 receptors in the bladder. Because cardiomyocytes possess γ_1 adrenaline receptors, the presence of catecholamines can promote myocardial contraction. This constricts and increases heart rate, thus increasing the amount of blood the heart outputs, and some of this is called beta-adrenergic receptor blockade. Beta blockers work by blocking the action of γ_1 -adrenergic receptors, thus slowing the heart rate. Smooth muscle cells possess γ_2 -adrenergic receptors, which react with catecholamines in the presence of these receptors. It can cause smooth muscle relaxation, and agonists targeting γ_2 -adrenergic receptors can be used for asthma. Treatment.



Based on the above description, which of the following is true (T) and which is false (F)?

- (A) Just because a drug works on one type of receptor does not mean it will work on another type of receptor.
- (B) γ -adrenergic receptor blockers are effective for any type of muscle.
- (C) γ -adrenergic receptors must be in the cytosol to affect contraction and relaxation
- (D) The protein chemical structures of γ_1 and γ_2 adrenergic receptors must have the same active site.

Narrative	(A)	(B)	(C)	(D)
T or F				

III. Multiple Choice Questions (7 points total)

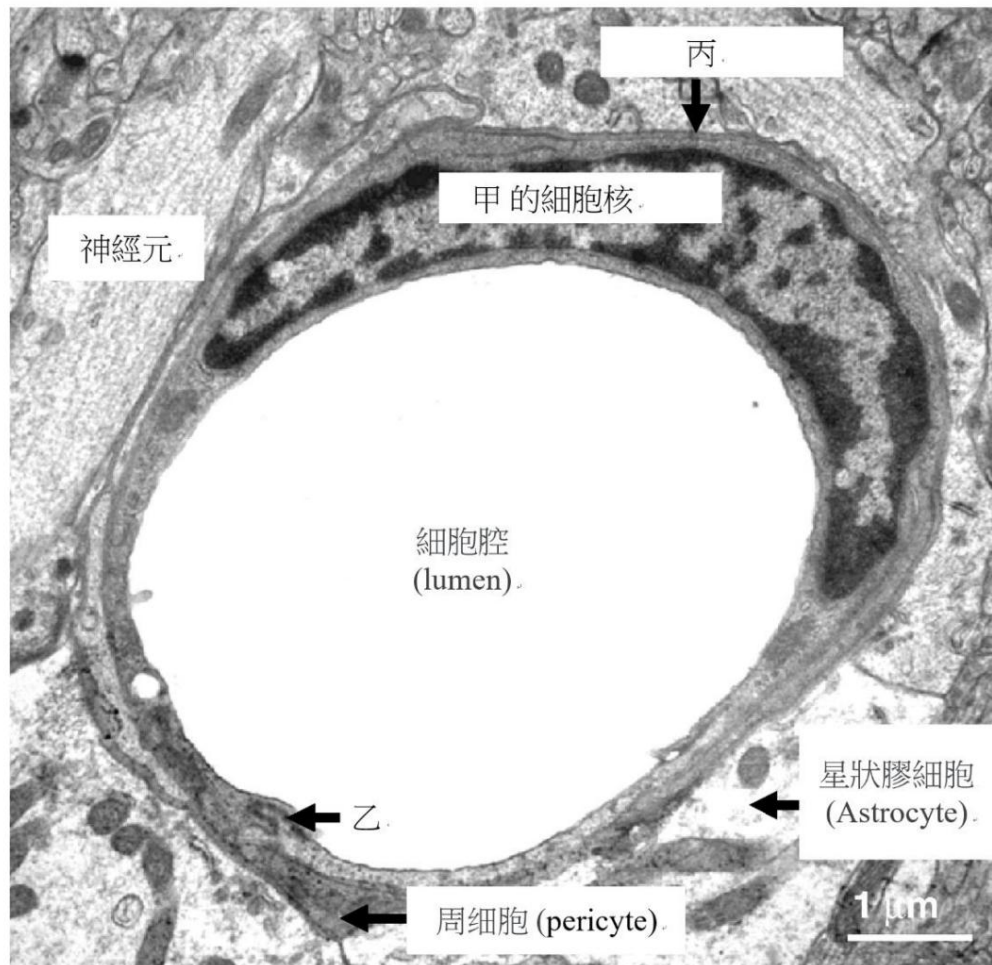
1. Multiple sclerosis (MS) is a disease of the central nervous system, meaning its lesions are located in the brain or spinal cord. It is generally considered an autoimmune disease, meaning that the individual's immune system attacks...

It attacks its own tissues. In multiple sclerosis, the immune system attacks the myelin sheath. However, when a person suddenly develops the disease...

However, MRI scans revealed blood-brain barriers in parts of the brain or spine.

Damage to the cerebral vascular barrier (BBB) allows T cells to enter and destroy the myelin sheath. The image below shows a penetrating penetration pattern at the cerebral vascular barrier.

Electron microscope image.



(1) At index A, what type of cell is most likely? (2 points)

- (A) Epithelial cells
- (B) Endothelial cells
- (C) Schwann cell
- (D) Glial cells

(2) At point B, what is the most likely structure? (2 points)

- (A) Gap junctions
- (B) Tight junctions
- (C) Desmosomes
- (D) Semi-desmosomes

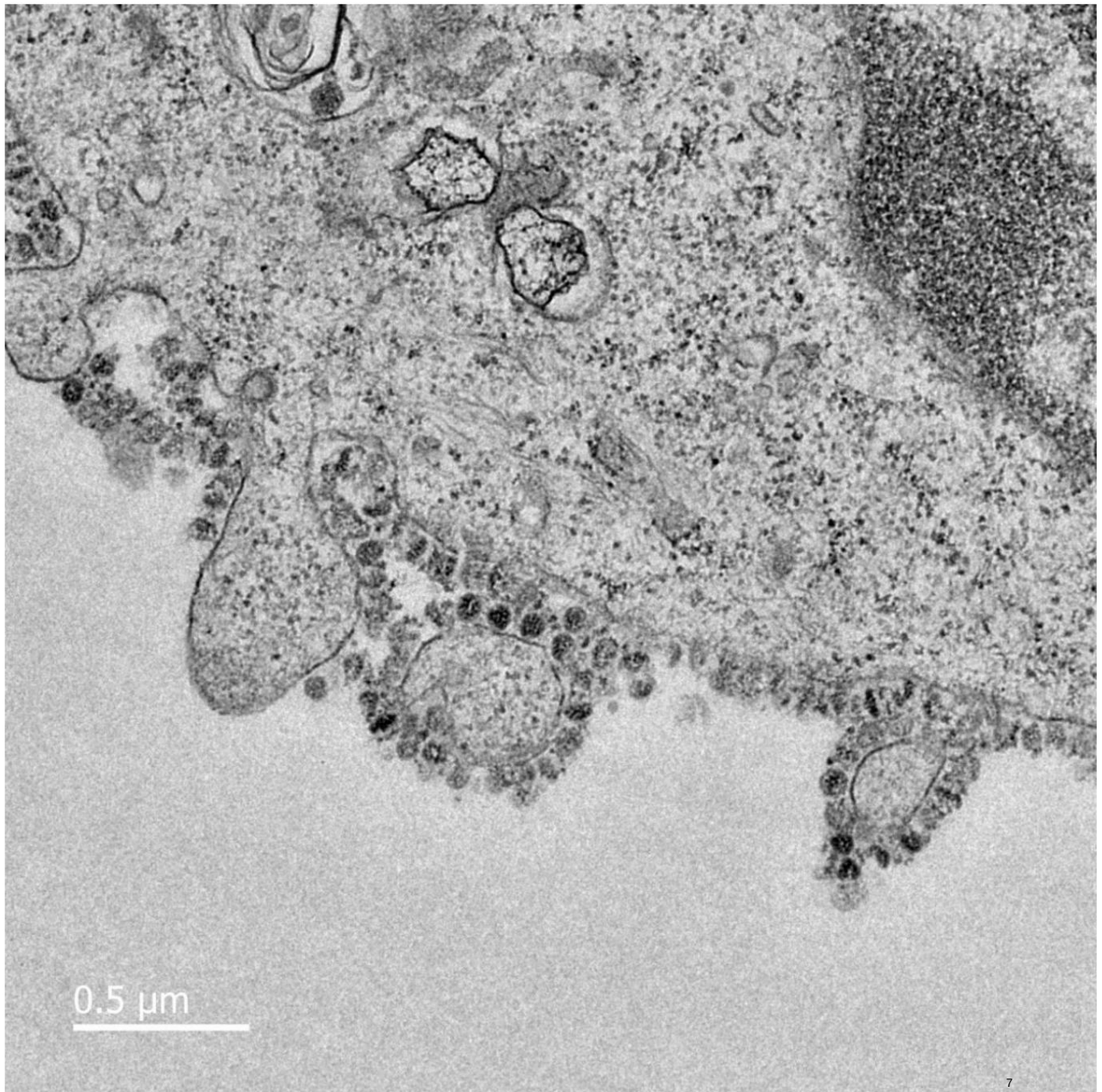
(3) Which of the following components is least likely to be present at index C? (3 points)

- (A) Laminin
- (B) Integrins
- (C) Collagen
- (D) Keratin

III. Question Sets (Total 10 points)

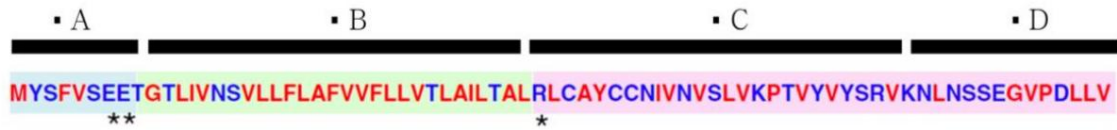
1. The image below is a thin-layer electron micrograph of a virus. It shows a portion of infected cells growing in a culture, with viral particles being released from the cell surface. Which of the following viruses is most likely the virus in question? (2 points)

- (A) Adenovirus
- (B) Norovirus
- (C) Coronavirus
- (D) Aquamavirus



2. Xiaoming decoded the structure of one of the proteins in the above-mentioned virus and obtained the following sequence diagram. Please answer the following questions. (Fill in the most correct region location in the answer, such as A, B, C, or D) (2 points per blank, 8 points in total)

point)



氨基酸性質：疏水性（紅色），親水性（藍色），極性，帶電（星號）。

Question Number	question	Answer
(1) N-terminal		
(2) C-terminal		
(3) Transmembrane region		
(4) Viral receptor binding region		

IV. Matching Questions (23 points in total)

1. Match the following to its definition: (2.5 points per blank, **15 points total**)

- a. Reproduction without gamete fusion (no parents required) b. A series of events that a cell undergoes during growth and division.
- c. The process by which a cell divides into two new "daughter cells" d. Each chromosome consists of two identical "sister strands" e. The region where each pair of chromatids attach f. Long, tightly wound, continuous segments of DNA, whose genetic information is passed down from generation to generation

proper nouns	definition (Please enter the code)
1. Cell division	
2. Asexual reproduction	
3. Chromosome	
4. Centromere	
5. Chromatid	
6. Cell cycle	

2. T cells: These are a type of lymphocyte, mainly including cytotoxic T cells and helper T cells.

Helper T cells, regulatory/suppressor T cells, and memory T cells

There are 4 types of T cells (memory cells). Clusters of differentiation (also known as differentiation groups, CDs) refer to...

These are used to identify cell surface molecules that serve as immune antigens. The total number of CD molecules in humans is 371.

Please correctly match the following four CD4 molecules for T cells. (2 points per blank, 8 points total)

Question Number	Types of T cells	Answer	Option CD	Molecular Types
example	B cells	AyG	A	IgM
(1)	Cytotoxic T cells (cytotoxic T cell)		B	CD3
(2) Helper T cells			C	CD4
(3)	Regulation/inhibition of T cells (regulatory / suppressor T cell)		D	CD8
(4) Memory T cells			AND	CD25
			F	CD45
			G	CD61
			H	CD117

V. Short Answer Questions (25 **points total**)

1. Please compare prokaryotic fermentation, aerobic respiration, and anaerobic respiration.

Differences and similarities between (anaerobic respiration). (10 points)

	Fermentation	Aerobic respiration	Anaerobic respiration
Capacity efficiency (Please fill in high or low)			
The presence or absence of an electron transport chain			
Ultimate recipient of electrons			
Site of action			
In addition to ATP, final products Object (please give any example) (Yes, points will be deducted for filling in too many or wrong answers.)			

- 2.** Epigenetics refers to the alteration of genome-related functions through certain mechanisms without changing the DNA sequence. These changes can be preserved through cell division and may last for several years.

Generation. Current research has revealed numerous mechanisms contributing to epigenetics. Please provide two examples and briefly describe them. (6 points)

- 3.** According to the endosymbiotic theory, the chloroplasts of eukaryotic cells originated from early cyanobacteria. Structural evidence supports this.

Currently, chloroplasts have two membranes, with the inner membrane folding inward to form a chloroplast disc, which is responsible for the light-dependent reactions of photosynthesis.

The part of the chloroplast that is located in the chloroplast. However, scientists discovered that the chloroplasts of the supergroup Chromalveolata have four membranes when studying them. Please provide a reasonable explanation. (4 points)

- 4.** Please briefly describe plasmodial slime molds and cellular slime molds.

The main differences between the two. (5 points)

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Theoretical Assessment Test Questions

Part Two

Student Code



Precautions:

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3. When answering the questions, please make sure the number on the cover of the test paper matches your test number.
4. Please answer the questions on the test paper using a blue (black) pen, and please keep the test paper clean.
5. You may write your answers on the back of the questions, but please indicate the question number.

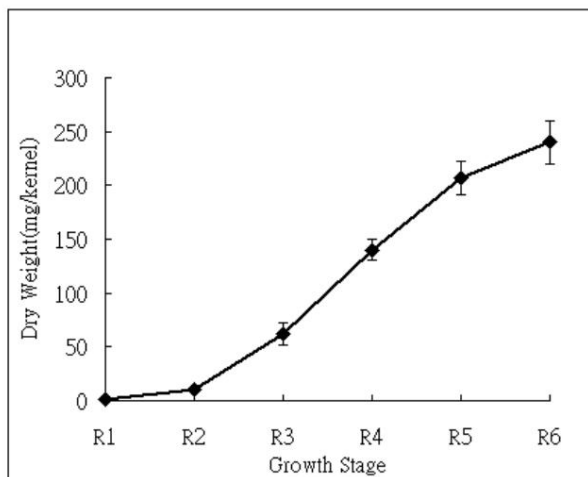
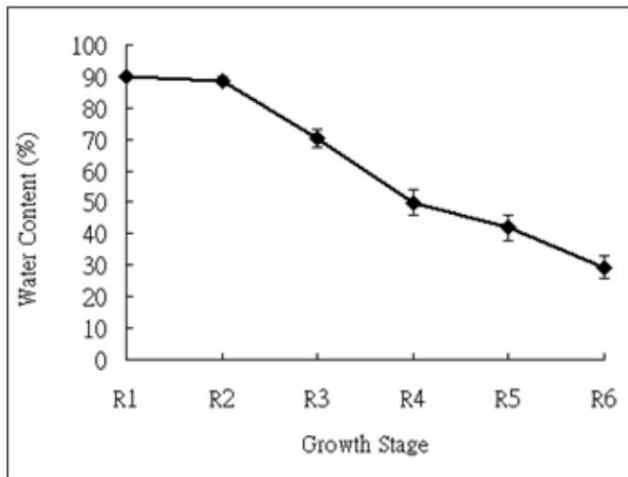
Part Two: (100 points)

The following are topics related to plant physiology. Please answer the questions according to the descriptions for each topic.

I. Crop scientists divide the development process of maize kernels (seeds) into R1-R6 stages (Richie & Hanway, 1989), as shown in the table below:

Grain development status	at developmental stages
R1	silking (silk drawing period)
R2	blister (during grain formation, the endosperm is clear and viscous)
R3	milk (during the milk-ripening stage, the endosperm becomes milky and then pasty).
R4	dough (waxy stage, endosperm is waxy)
R5	dent (Dent stage, the surface of the grain tip shows a small indentation)
R6	Physiological maturity (full maturity, harvest period)

Xiaoming tested the water content and dry weight of grains at each developmental stage, as shown in the following figure:



Answer the following questions based on this information:

1. In which stages of the corn kernels are (1) carbohydrate filling capacity and (2) starch content higher, respectively, during the R3 and R5 stages?

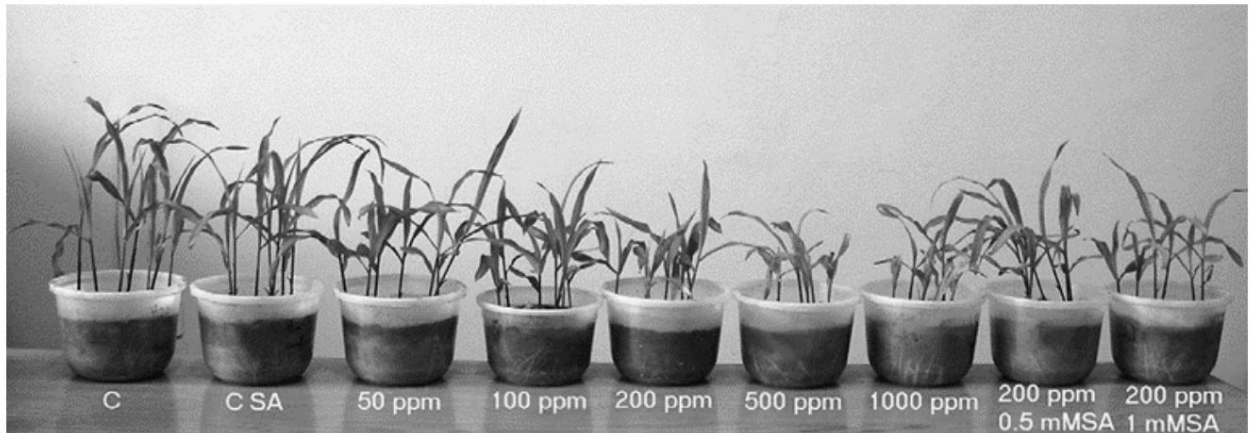
What is the reason? (10 points)

2. In corn kernels, which stage has a higher content of the plant hormone ABA, R3 or R5? What is the reason? (2) Water potential

Which has a higher absolute value, the matrix potential, in period R3 or R5? What is the reason? (10 points)

II. The following figure shows the growth of corn seedlings under treatment with herbicide (Clethodim herbicide) at 50-1000 ppm. SA represents...

Salicylic acid treatment (Note: the molecular weight of salicylic acid is 138), C is the control group.



The table below shows the results of Xiaoming's treatment with herbicides and salicylic acid to stabilize the intermediate products MDA and H₂O₂ of lipid peroxidation in leaves.

Test results:

deal with	MDA%	H ₂ O ₂ %
C	100	100
C+SA	111	112
Herbicide 50 ppm	144	101
100 ppm	93	123 $\ddot{y}$
200 ppm	169 $\ddot{y}$	136 $\ddot{y}$
500 ppm	220 $\ddot{y}$	150 $\ddot{y}$
1000 ppm	272 $\ddot{y}$	163 $\ddot{y}$
herbicide, 200 ppm + SA	100	102

Note: The probability of a significant difference from the control group is: $\ddot{y}$: $p < 0.05$, $\ddot{y}$: $p < 0.01$

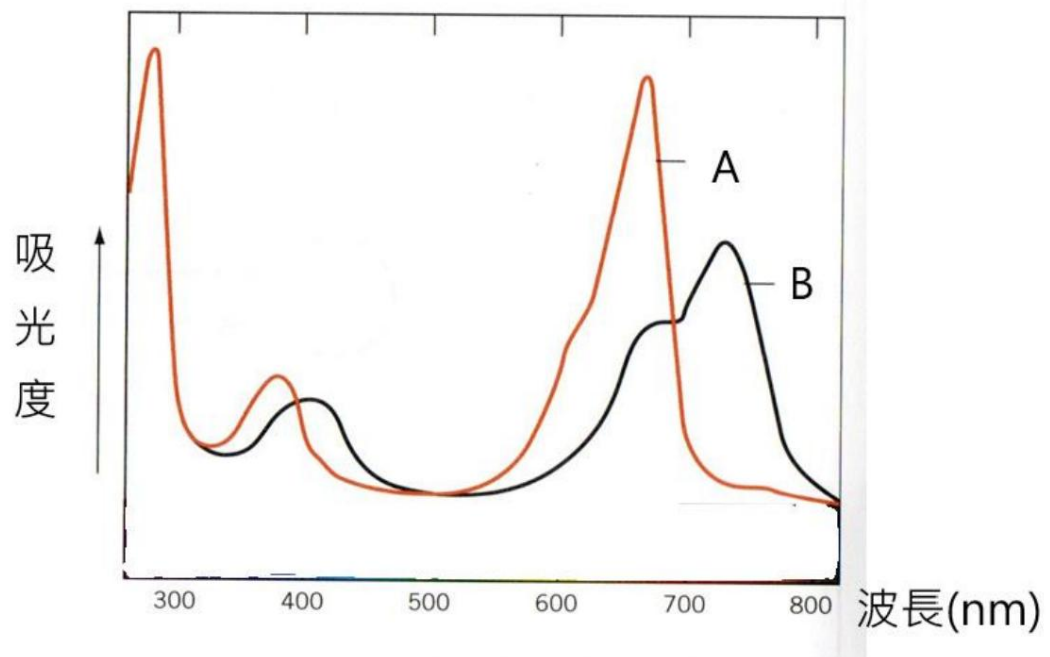
Answer the following questions based on this information:

1. How to prepare a 1000 ppm herbicide solution? How to prepare a 1.0 mM SA solution? (10 points)

2. In this experiment, what is the main damage that herbicides cause to corn? What is the reason? (2) What is the function of SA?

Why? What is the reason? (10 points)

III. The following diagram shows the absorption spectrum of phytochrome (10 points).

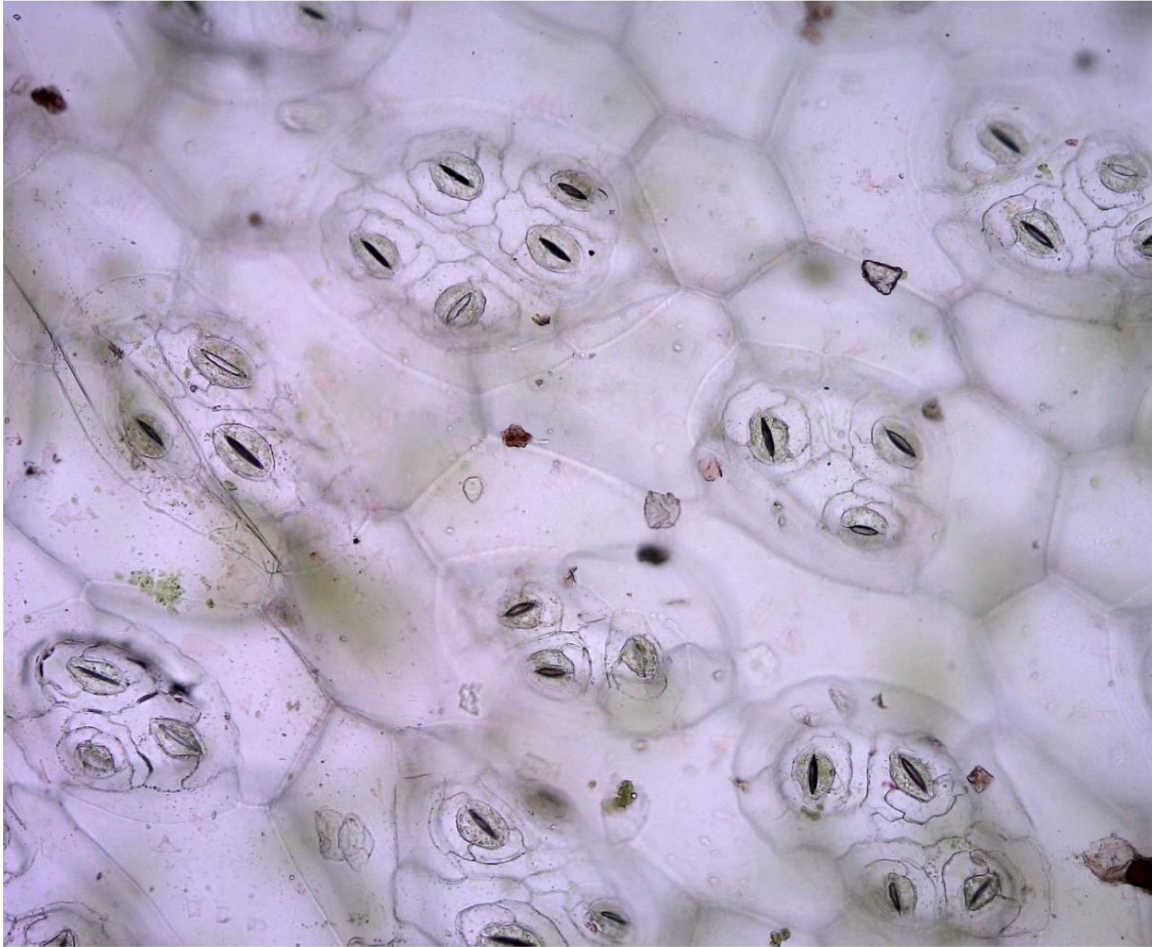


1. What are A and B in the diagram? What is the reason for your judgment?

2. What are the physiological functions of A and B?

IV. The following are topics related to plant morphology. Please answer the questions according to the descriptions for each topic.

1. In the lower epidermis of *Begonia semperflorens*, stomata are distributed in groups of 3-5, as shown in the figure below.



(1) Based on this observation, what ecological advantages does this distribution pattern offer to *Begonia semperflorens*? (10 points)

(2) Based on the above hypothesis, please design an experiment to verify or refute it. (15 points)

2. There are many varieties of poinsettias. The following image shows the flower organ morphology of a common variety (A) and another variety (B).



(Figure A) Common poinsettia and a magnified view of its central area.



(Figure B) Another variety of poinsettia (Peach Delight)

(1) Describe the morphological characteristics of the floral organs of the two varieties above. (10 points)

(Figure A) _____

(Figure B) _____

(2) Infer which floral organ morphology was the focus of selection and cultivation for variety B in Figure B ? (5 points) What is the possible operational principle behind this selection? (10 points)

point)

Key points for selection and training: _____

Selection and cultivation operation principle: _____

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Theoretical Assessment Test Questions

Part Three

Student Code



Precautions:

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4. Please answer the questions on the test paper using a blue (black) pen, and please keep the test paper clean.
5. You may write your answers on the back of the questions, but please indicate the question number.

Part Three: (100 points)

1. Have you heard of the ketogenic diet? The ketogenic diet has recently been in the spotlight, with many people using it to lose weight, but is it really okay? The ketogenic diet is a high-fat, high-protein, low-carbohydrate diet.

This diet forces the body to burn fat instead of carbohydrates. However, the nerve cells in the brain can only...

Using glucose as an energy source, the brain's nerve cells require alternative energy sources when glucose is scarce. The body uses acetyl-CoA, a small molecule produced by the oxidation of fats and proteins in the liver, to generate large amounts of ketone bodies, including γ -hydroxybutyrate and acetoacetate. These liver-produced acetoacetate and γ -hydroxybutyrate are then transported via the bloodstream to brain nerve cells for metabolism. Furthermore, it has been shown that γ -hydroxybutyrate can activate a series of intracellular signaling pathways through specific G protein-coupled receptors (GPRs), such as GPR41 or GPR109; however, the receptors for acetoacetate and their metabolic processes or regulation require further investigation.

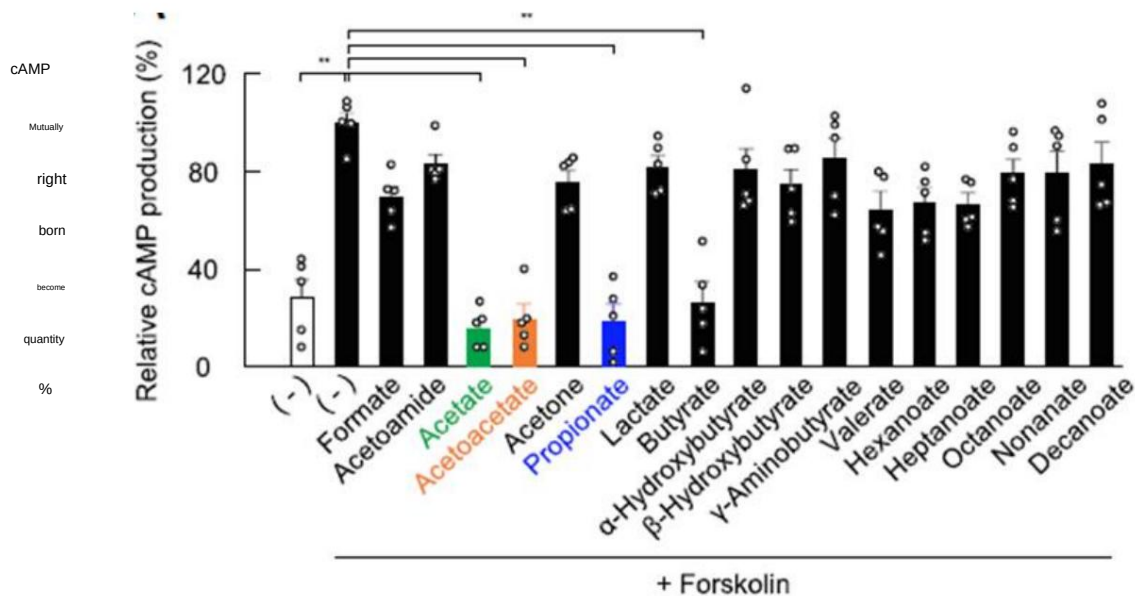
The following is an article published in the Proceedings of the National Academy of Sciences of the United States of America in 2019 by Miyamoto et al. (Miyamoto et al., 2019. Proc. Natl. Acad. Sci. USA 116, 23813–23821), in which they argue that GPR-43 may be involved in the process.

It plays an important role in [the process]. Please answer the following questions based on the provided chart.

Figure A shows the intracellular cAMP levels in a large number of GPR-43 (Flp-in GPR43 T-REx HEK293) cells after different treatments. These cells were administered IBMX, a phosphodiesterase inhibitor, before drug administration.

Treatment with phosphodiesterase inhibitor for 24 hours, followed by the addition of various monocarboxylic acids,

The concentration of intracellular cAMP was determined using ELISA.



(Figure A)

1. Please explain why you should first submit your application to IBMX for processing within 24 hours. (3 points)

2. Please explain the role of forskolin in this experiment. (3 points)

3. G-coupled proteins can be classified into Gs or Gi based on their subsequent effects. Which type is GPR-43? Please explain.

Reason (3 points)

4. Please summarize the main research findings presented in this figure in one sentence (within 10 characters in Chinese, for a single carboxylic acid).

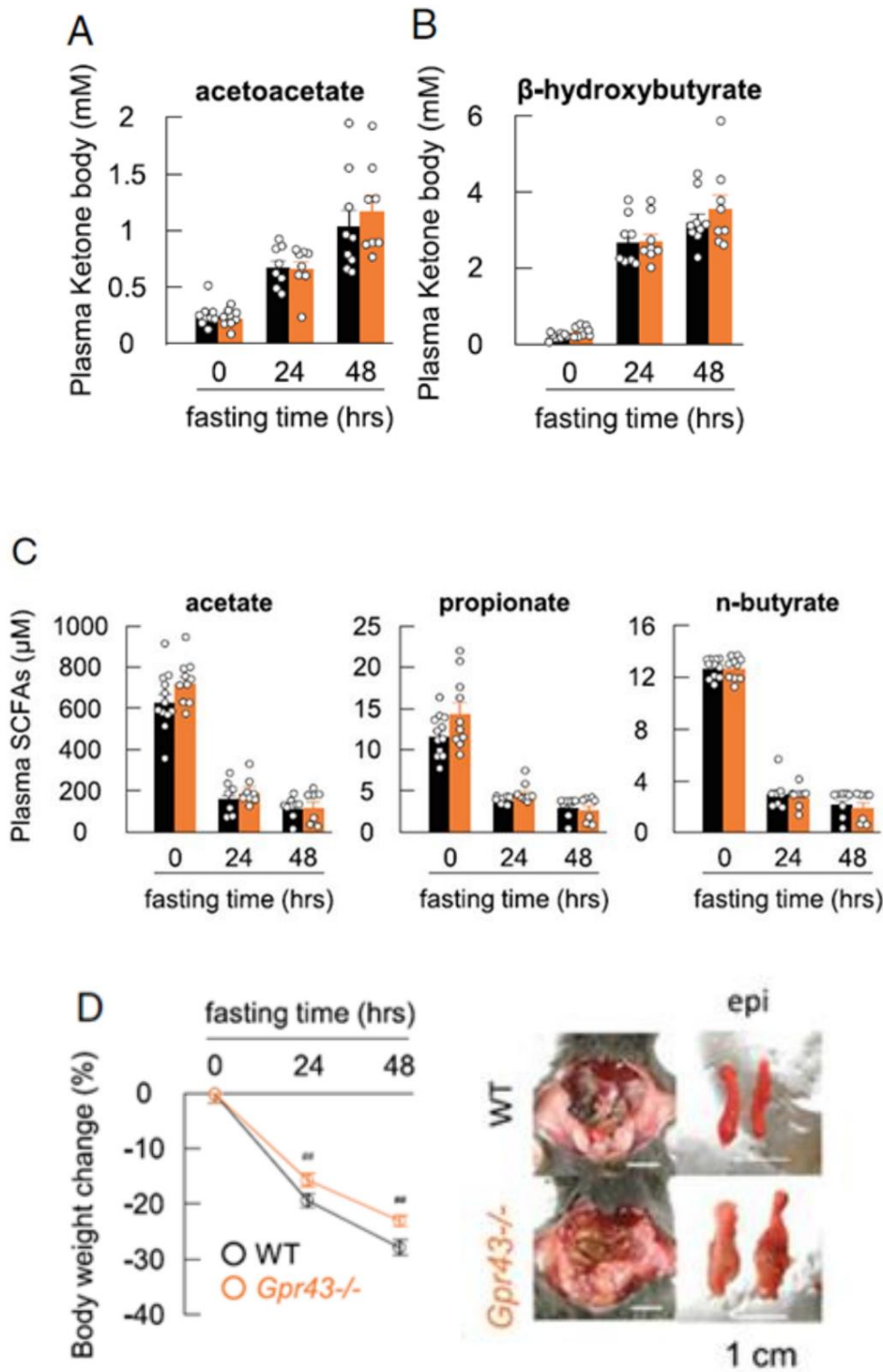
The English name is considered as one word; deduct one point for each extra word. (6 points)

Second, the authors then used GPR43 knockout mice (*Gpr43*^{-/-}) to analyze the effects of different conditions, including normal and fasting states.

Afterwards, the levels of plasma ketone bodies and short-chain fatty acids (plasma SCFA) in the blood of mice were measured.

Light-colored mice represent GPR43 knockout mice (*Gpr43*^{-/-}), and dark-colored mice represent wild-type mice. Please refer to the diagram below.

Answer the following questions.



Part Three

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1. In a fasting state, does GPR43 primarily interact with ketone bodies or short-chain fatty acids? (2 points) Why? (3 points)

2. Please summarize the conclusion of Figure D in one sentence (in Chinese, within 22 characters; one point will be deducted for each extra character) (5 points)

Third, when performing animal dissections, it is not enough to simply read from the textbook, follow the diagrams, or skim through the lecture notes.

More importantly, we should make judgments, inferences, and verifications based on the clues we observe.

1. Please list all the physical symmetries of animals, and using an animal from a specific phylum as an example, explain how to determine its symmetry during dissection. (10 points)
2. Please explain how to determine the head and tail of an animal during dissection. (5 points)
3. Please explain how to determine the sex and sexual maturity of an animal during dissection, and provide one example each for vertebrates and invertebrates (10 points).

IV. The image on the right shows the sciatic nerve after stimulation.

A diagram showing the generation of an action potential (the value that generates the action potential)

The value is -55mV , and the action potential is mainly related to voltage sensitivity.

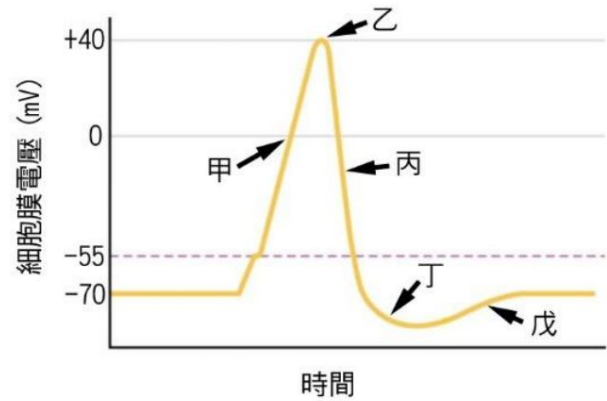
Voltage-dependent Na^+ channel (VNa ; please use VNa instead in the

following answers) and electrical...

Related to pressure-sensitive potassium ion channels

(voltage-dependent K^+ channel; Answer below)

Please use VK instead. Please answer the following questions respectively:



1. Why doesn't the rise in cell membrane voltage remain at 0 (position "A" in the diagram) during the depolarization phase of the action potential?

Is it continuing to rise? (5 points)

2. If VNa opens when the cell membrane voltage exceeds the threshold, why does the voltage reverse downwards at point "B"?

What? (3 points)

3. If VNa will open due to cell membrane voltage exceeding the threshold, the roles of VNa and VK at point "C" are as follows:

What? (3 minutes)

4. When the cell membrane voltage is below -70mV , theoretically, the cell membrane voltage is already below the activation thresholds for VNa and VK .

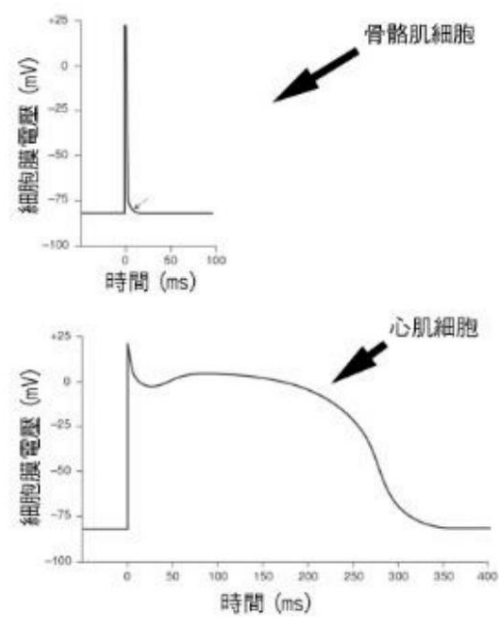
Why does cell membrane voltage hyperpolarization still occur (as shown at position "D" in the figure)? (4 points)

5. Why does the cell membrane voltage rise slightly (as shown in position "E" in the figure) after hyperpolarization and then return to rest?

What about the membrane voltage of -70mV ? (5 points)

V. The right figure shows the action electrical activity of skeletal muscle cells and ventricular muscle cells.

Please compare the differences between the two positional graphs. (5 points)



VI. Questions and Answers

1. The diagram below is a comparative illustration of vertebrate embryonic development drawn in 1892 by Canadian evolutionary biologist George Romanes, based on a work by German biologist Ernst Haeckel. This diagram is frequently used to support the theory of embryonic evolution.

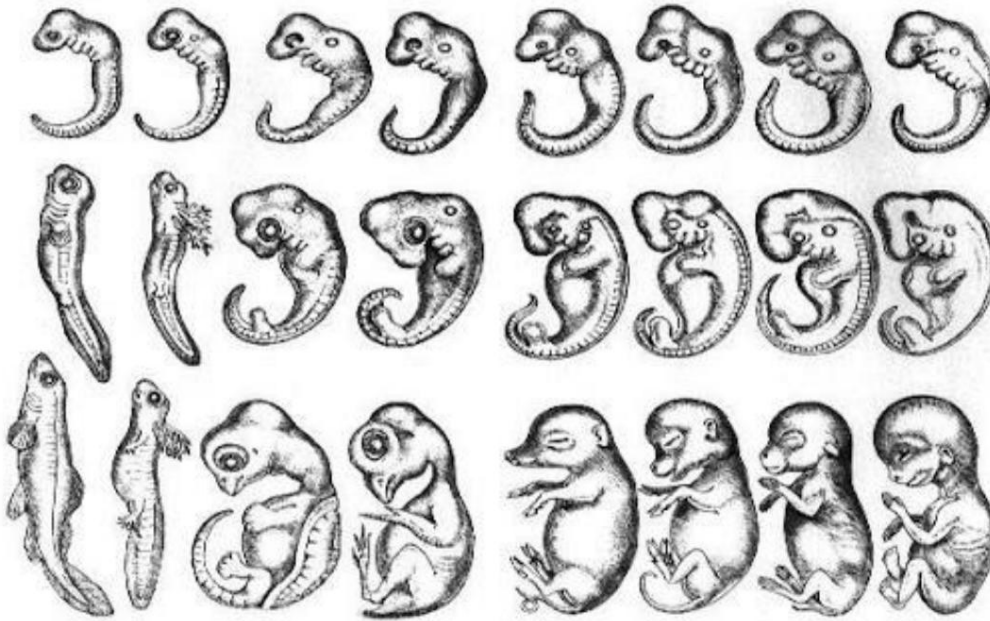
The theory of recapitulation. The so-called embryonic evolution recapitulation theory refers to the theory that "an animal develops from an egg into a mature animal..."

The morphological changes observed during fetal development reflect the evolutionary process of an animal from its ancestors to its descendants. For example, humans...

Human embryos still possess gill slits, thus the process of human embryonic development reconstructs the "evolutionary history of vertebrates."

This theory has been refuted and disproven in modern times. Based on the principles of evolution and embryonic development, what is your opinion on this?

Why was this theory rejected? (12 points)



2. Figure 2-1 illustrates divergence, evolutionary conservation, parallelism, and convergent evolution.

A schematic diagram illustrating the definition of convergence. Figure 2-2 shows the phylogenetic relationships of a group of sea urchins and their development from larvae to adults.

The process of reproduction. Based on the information in these two images, what is the role of sea urchin larvae with long arms that can feed on their own within this type of sea urchin?

Which type of AD does the evolutionary pattern belong to? Why? (5 points)

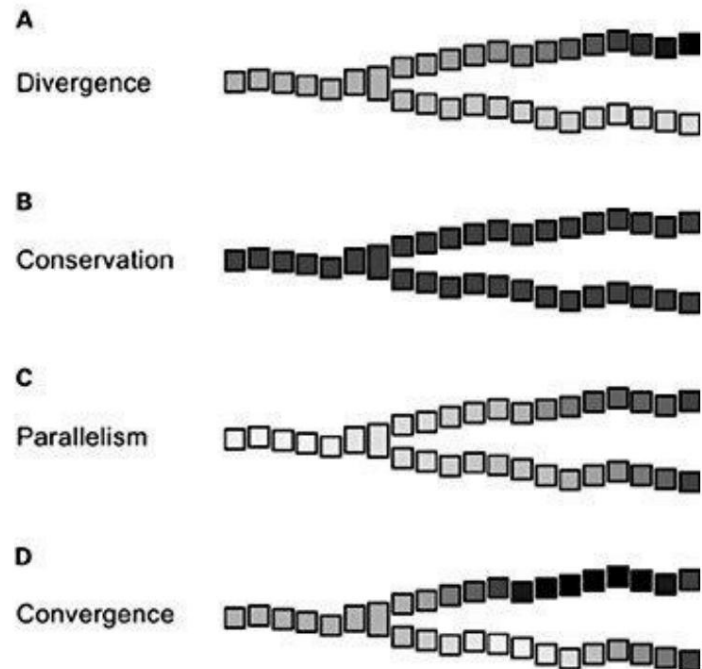


Figure 2-1.

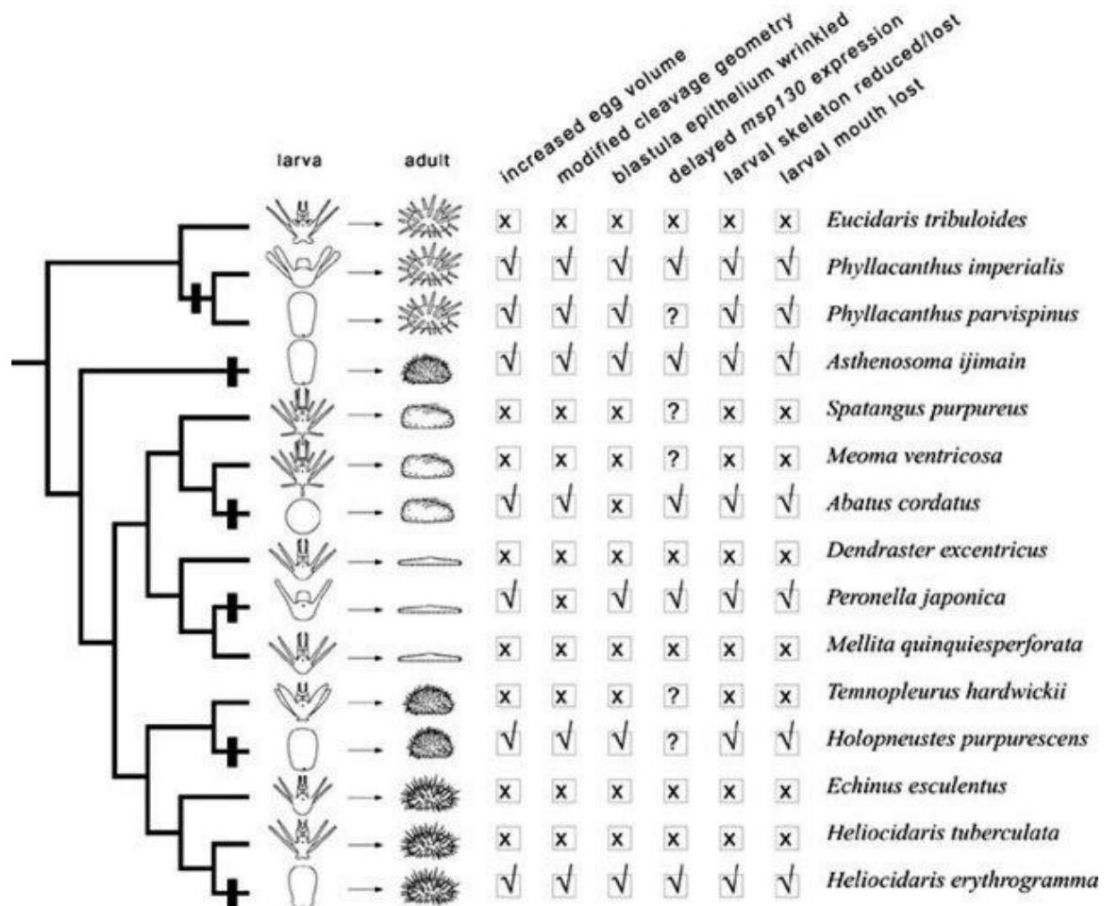


Figure 2-2.

3. In the evolutionary history of the animal kingdom, is the "larval" developmental stage more evolutionarily significant than the "adult" stage?

The existence of the "adult" or the "larva" appearing earlier are two completely opposing hypotheses. Let's assume you now want to...

If we use adults and trochophore (larvae of the phylum Mollusca) as research subjects, then please ask...

Which type of genetic information do you think would be best used to address this issue? Why? (8 points)

- (A) plasmid
- (B) Mitochondria
- (C) Microsatellite
- (D) Transcriptome
- (E) Ribosome

2020 Republic of China Biology Olympiad National Team

Selection Camp

Theoretical Assessment Test Questions

Part Four

Student Code



Precautions:

1. This section of the test consists of 6 pages (including the cover) and is worth 100 points.
2. Please check for missing pages. If any pages are missing, please inform the test administrator immediately.
3. When answering the questions, please check if the cover of the test paper matches your test number.

4. Please answer the questions on the test paper using a blue (black) pen, and please keep the test paper clean.
5. You may write your answers on the back of the questions, but please indicate the question number.

Part Four: (100 points)

I. Short Answer Questions (80 points total)

1. The cell division cycle can be divided into four phases: G₁, S, G₂, and M. In the G₁ phase, the cell...

Growth occurs during this phase, with an increase in size and organelles; in the S phase, DNA replication takes place; in the G₂ phase, organelles increase, and cell division occurs.

Required proteins to prepare for cell division; M phase, where nuclear division (mitosis) and cytoplasmic division occur, cell...

Divided into two parts. The initiation and termination of each stage of the cell division cycle are regulated by cyclin-dependent kinases

(CDKs). CDKs must bind to cyclins to be active.

Different cyclins combine with CDK at different times, activating CDK and determining the specificity of CDK receptors.

CDK combines with S-cyclin to form S-cyclin-CDK; CDK combines with M-cyclin to form M-cyclin-CDK. When various conditions, such as cell size, growth factors, and nutrients, are met, S-cyclin-CDK is activated, allowing the cell to enter the S phase and initiate DNA replication. Once DNA replication is successfully completed, other conditions...

When everything is normal, M-cyclin-CDK will be activated, causing the cell to enter the M phase, initiating mitosis and subsequent cell division.

Cytokinesis (as shown in the image below).

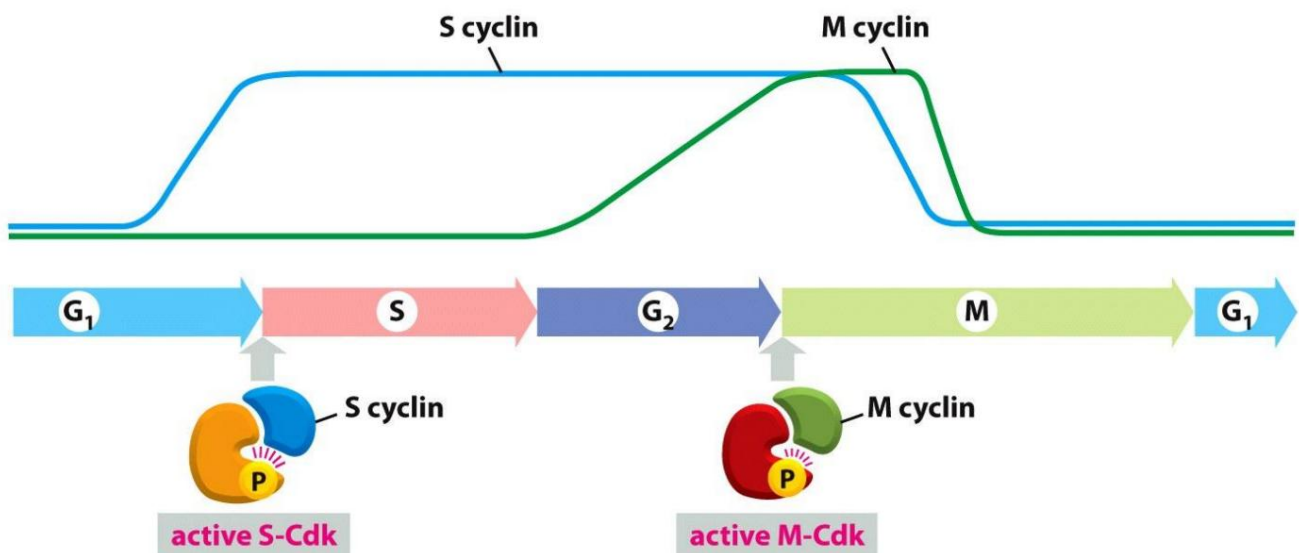


Figure 18-10 Essential Cell Biology 3/e (© Garland Science 2010)

The activities of S-cyclin-CDK and M-cyclin-CDK are turned off before the cell completes division (M phase), allowing the cell to migrate.

The S phase begins, and the CDK re-enters the G₁ phase. Therefore, CDK activity continues from the S phase to the M phase, while no CDK activity occurs in the G₁ phase.

The activity is inhibited because there is no synthesis of S-cyclin or M-cyclin, and there is also the presence of CDK inhibitor (CKI), which can inhibit the activity of S-cyclin-CDK and M-cyclin-CDK.

During the G₁ phase, certain proteins bind to the DNA replication origin, forming the "pre-replicative complex" (pre-RC). Only replication origins with pre-RC can initiate DNA replication.

During replication, the pre-RC disappears and transforms into the post-replicative complex (post-RC), which does not have the function of initiating DNA replication. Furthermore, the formation of the pre-RC is inhibited by CDKs.

For the replication origin to be formed, pre-RC cannot be formed unless S-cyclin-CDK or M-cyclin-CDK activity is present.

Based on the background information above, please answer the following questions:

(1) Besides the G1 period, can the G2 period form a pre-RC? Why? (10 points)

(2) Experiments have shown that when G1 phase cells are fused with S phase cells, the G1 phase cells immediately enter the S phase, thus...

What? (10 points)

(3) If cells in G2 phase and cells in S phase are fused, will the cells in G2 phase immediately enter S phase? Why?

(10 points)

(4) Before the end of cell division, shutting down the activity of S-cyclin-CDK and M-cyclin-CDK has an effect on the next cell division cycle.

What is its function? (10 points)

(5) The seemingly complex relationship between pre-RC, post-RC, CDK, and DNA replication during the cell division cycle plays a crucial role in regulating cell division.

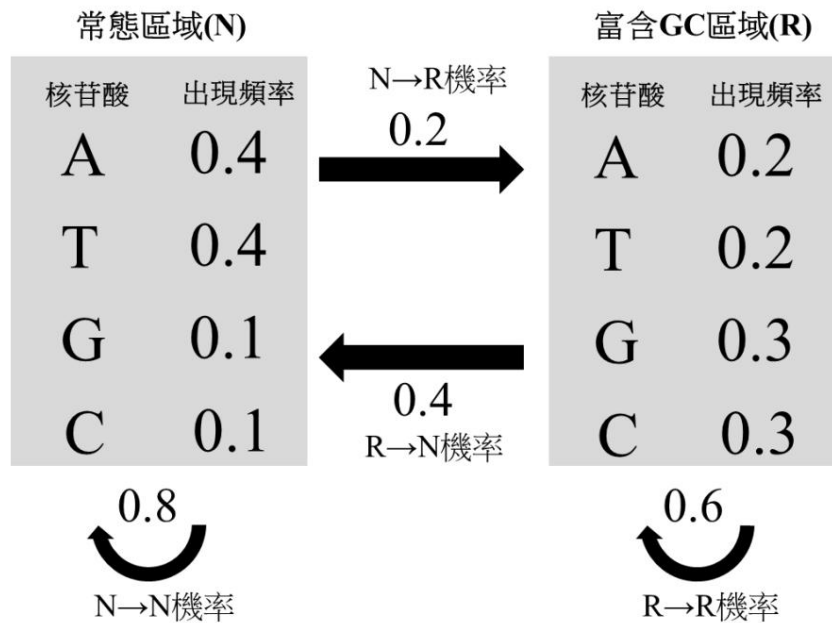
What are the special functions or biological significance of the cell division cycle? (10 points)

2. Hidden Markov Models (HMMs) are commonly used in the fields of bioinformatics and genomics.

Hidden Markov Models (HMMs) are widely used in major bioinformatics databases such as Ensembl. The fundamental principle of HMMs is that each nucleotide in a DNA sequence to be identified is divided into a normal region (N) and a GC-rich region (R). The prediction calculation method is as follows:

$$\text{Probability (Sequence)} = [\text{Probability (Frequency of Occurrence)}] \times [\text{Probability (Probability of Conversion Between NR)}]$$

If we find the frequency of nucleotides in a sequence in normal and GC-rich regions as shown in the figure below, please answer the following questions.



- (1) Based on the principle of the hidden Markov model, please calculate the probability of the TGC sequence appearing in the NRR hidden state (IV).

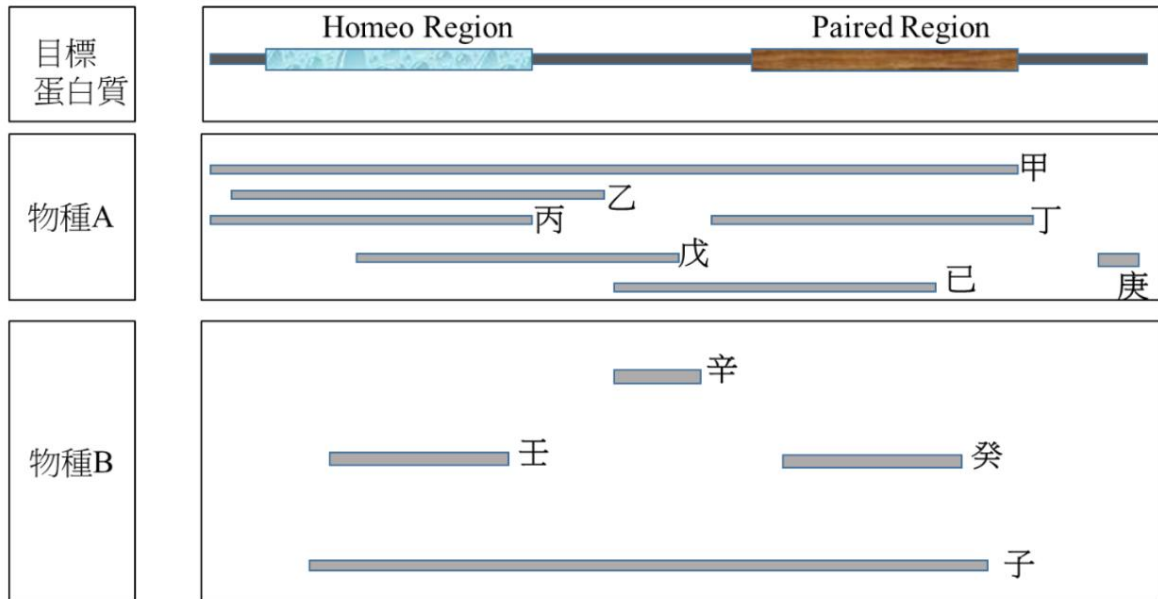
Rounding down to the fourth decimal place? (10 points)

- (2) If a DNA sequence is divided into six regions, when the first nucleotide is in the normal region (N), there may be multiple

How many hidden groups are there? (6 points)

3. NCBI offers various sequence comparison services. After inputting a sequence into a database, researchers can find the most similar sequence fragments based on the selected database type. Suppose we obtain an unknown target protein...

The sequence, after being sorted through a database, yields the result shown in the image below. Please answer the following questions.



(1) Based on the above diagram, which species has a relatively high similarity to the target protein? (3 points)

(2) Which sequence has the lowest similarity? (3 points)

(3) How many sequences in the above diagram are similar to the paired region of the target protein? (3 points)

(4) In the comparison results between species A and B, which sequence has the highest similarity to the target protein? (5 points)

II. Matching Questions (20 points total)

1. DNA sequencing began in the 1970s, laying the foundation for the development of various fields in contemporary life sciences. The vast amount of sequence data accelerated the development of bioinformatics algorithms. Please match the following descriptions of various sequencing methods to the appropriate options.

The corresponding ordering method (each option may be used more than once).

(A) Sanger sequencing (B) Maxam-Gilbert sequencing

(C) Massively parallel signature sequencing (D) Single molecule fluorescence sequencing

(E) Nanopore sequencing (F) Illumina sequencing (G) Ion semiconductor sequencing (X) DNA nanoball sequencing (I) Single molecule real-time sequencing

Example: The ordinal method developed by Dr. Frederick Sanger in 1977. Answer: A

_____ (1) This sequencing method uses synthetic sequencing. The principle is to attach a fluorescently labeled deoxyadenosine triphosphate to the 3' end of a DNA fragment of 100-200 base pairs using terminal deoxynucleotidyl transferase.

The poly-A tail is used for subsequent nucleotide identification. (5 points)

_____ (2) This sequencing method is also known as chemical sequencing. The sequencing process involves radiolabeling the 5' end of the DNA sequence fragment.

Note that DNA purification is required. (5 points)

_____ (3) This sequencing method benefits from the development of zero-mode waveguide technology. Each zero-mode waveguide aperture carries a DNA polymerase and a DNA fragment, which are used for sequencing and simultaneously read the fluorescent signal

carried by the inserted bases. (5 points)

_____ (4) The materials required for this sequencing method are single-stranded DNA template, sequencing primers, deoxyribonucleoside

triphosphates (dNTPs), dideoxyribonucleoside triphosphates (ddNTPs), and DNA polymerase. It was the most widely used sequencing method before the year 2000. (5 points)

2020 Republic of China Biology Olympiad National Team

Selection Camp

Theoretical Assessment Test Questions

Part Five

Student Code



Precautions:

1. This section of the test consists of 9 pages (including the cover) and is worth 100 points.
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3. When answering the questions, please check if the cover of the test paper matches your test number.
4. Please answer the questions on the test paper using a blue (black) pen, and please keep the test paper clean.
5. You may write your answers on the back of the questions, but please indicate the question number.

Part Five: (100 points)

I. Besides affecting human health, some diseases, such as Severe Acute Respiratory Syndrome (SARS) which emerged in 2003 and the current novel coronavirus pneumonia (COVID-19), can also have a significant impact on human social and economic activities. (Total 29 points)

_____ 1. Which of the following human diseases are caused by viruses? (Multiple choice, 1 point each, total 5 points)

- (A) Malaria
- (B) Measles
- (C) Pulmonary tuberculosis
- (D) Sleeping sickness
- (E) Smallpox

_____ 2. Many human diseases are transmitted from animals to humans; these diseases are called zoonotic diseases.

(zoonoses). Which human diseases are zoonotic diseases, including those that, although now transmissible between humans, originated from other animal species? (Multiple choice, 1 point per option, total 5 points)

- (A) AIDS
- (B) Malaria
- (C) Measles
- (D) Smallpox
- (E) Dengue fever

_____ 3. Many important infectious diseases originated from pathogens carried by bats. Which of the following human diseases, as far as we know, are caused by pathogens carried by bats? (Multiple choice, 1 point for each option, 5 points in total)

- (A) Ebola
- (B) Middle East Respiratory Syndrome (MERS)
- (C) Rabies
- (D) Nipah virus-causing encephalitis
- (E) Hantavirus-causing hemorrhagic fever with renal syndrome)

4. Researchers use mathematical models to help understand the dynamics of infectious disease transmission among populations. One frequently used mathematical model is called SIR, where S stands for susceptible (individuals who may be infected), I for infected (individuals who are already infected), and R for recovered (individuals who have been infected but are no longer susceptible to infection). The SIR model is shown in the figure below:



This includes the chance of infection after contact with an infected person (I), and the speed of recovery after infection.

Rate. If individual mortality is disregarded, the changes in the quantities of S, I, and R over time can be expressed as:

$$\frac{dS}{dt} = N - \frac{A}{dt}$$

$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$\frac{dR}{dt} = \frac{B}{dt}$$

Where N is the population size

A and B should be filled in respectively: _____ (2 points each, 4 points in total)

5. Continuing from the previous question, if the number of infected individuals (I) remains constant and does not change over time, then:

$$\frac{dI}{dt} = \beta SI - \gamma I = 0$$

$$\beta SI = \gamma I$$

$$S\beta/\gamma = 1$$

This can be defined as the basic reproduction number (R_0), where R_0 is...

The higher the value, the more likely the number of cases will increase, and vaccination works by reducing S or γ .

This method is used to prevent the spread of disease. It assumes that all vaccinated individuals will not be infected, and that all individuals...

If individuals have equal opportunities to come into contact with each other, then calculate the minimum number of contacts required in a group based on the following conditions.

What percentage of individuals need to be vaccinated to prevent an increase in cases?

(1) $R_0 = 4$: (5 minutes) _____

(2) The higher the R_0 value, the higher the proportion of the population that needs to be vaccinated to prevent the spread of the disease.

High or low? _____ (2 points)

(3) Based on current knowledge, which of the following diseases has the highest R_0 value? _____ (Single choice)

(Question, 3 points)

(A) Smallpox

(B) Measles

(C) AIDS

(D) Ebola

(E) Novel Coronavirus Pneumonia (COVID-19)

II. The table below shows the standing biomass at three trophic levels in four different ecosystems.

Data on crop (kg/m²), gross productivity (g/m²/year), and net productivity (g/m²/year). (Total 36 points)

A ecosystem	Gross and net productivity of living biomass		
consists of	45.00	4500	2000
producers, primary	0.50	250	100
consumers, and secondary consumers.	0.01	60	12
B. Ecosystem	Gross and net productivity of living biomass		
Producers,	15.00	6500	2500
Primary Consumers,	3.50	480	200
Secondary Consumers	0.20	40	15
C Ecosystem	Gross and net productivity of living biomass		
Producers	5.00	1700	600
Primary Consumers	1.50	120	40
Secondary Consumers	0.05	8	3
D Ecosystem	Gross and net productivity of living biomass		
Producers	0.005	300	120
Primary Consumers	0.008	80	18
Secondary Consumers	0.001	5	2

1. If ecosystem A is a tropical rainforest, then ecosystem B is most likely a temperate grassland.

(temperate grassland), temperate coniferous forest, tundra

Which of the following is NOT a type of marine environment: tundra, tropical swamp, or pelagic ocean?

One? Why? (4 points)

2. If ecosystem A is a tropical rainforest, then ecosystem C is most likely a temperate grassland.

(temperate grassland), temperate coniferous forest, tundra

Which of the following is NOT a type of marine environment: tundra, tropical swamp, or pelagic ocean?

One? Why? (4 points)

3. If ecosystem A is a tropical rainforest, which of the following is most likely ecosystem D:

temperate grassland, temperate coniferous forest, tundra, tropical swamp, or pelagic ocean?

Why? (4 points)

4. In these four ecosystems, which ecosystem's primary producers produce the most dead organic matter?

How much dead organic matter does this ecosystem's primary producers produce annually? (4 points)

5. In these four ecosystems, which ecosystem has the largest energy flux in its detritus food web?

(Energy flux)? How to determine this? (4 points)

6. Among consumers at each trophic level in these four ecosystems, which ecosystem has the lowest trophic transfer efficiency between which trophic level? What is its trophic transfer efficiency?

(4 points)

7. In which of these four ecosystems do producers have the lowest biomass turnover?

What is the turnover time of its living biomass? (4 points)

8. Please draw the energy pyramid (or energy tower) of ecosystem D. (4 points)

9. Please draw the biomass pyramid for ecosystem D. (4 points)

III. Although Taiwan is a small island, it boasts over two hundred mountains exceeding 3,000 meters in altitude. Its complex topography and geological conditions, coupled with diverse climate variations, have resulted in high biodiversity and a wide variety of plant communities. Based on the results of a survey of 3,824 sample plots in Taiwan's forest vegetation, forests can be classified into 13 types. Figure 1 shows the climatic environment of the sample plots for each forest type, and Figure 2 shows the altitudinal distribution. (Image modified from Lin et al. 2020 Applied Vegetation Science). Please refer to Figures 1 and 2 to answer the following questions, filling in the forest type code. (Total 35 points)

1. Which forest type has the greatest temperature difference between winter and summer?

_____ (3 points)

2. Which forest type has the southernmost distribution range in Taiwan?

_____ (3 points)

3. Which forest type is mainly distributed in northeastern Taiwan?

_____ (3 points)

4. Which of the following options lists forest types whose climate has the most distinct dry season?

(1) Black Tea-Phoebe zhennan winter monsoon forest (2) Lithocarpus serrata-Euonymus decipiens forest (3) Phoebe bournei-Phoebe zhennan forest (4) Oak cloud forest

_____ (3 points)

5. Which forest type has its vegetation sample plots distributed at altitudes above 1800 m?

_____ (3 points)

6. Which of the following forest types has the largest potential distribution area in Taiwan? (1) Yushan Juniper Shrubland

(2) Cypress Cloud Forest (3) Castanopsis Forest (4) Ficus Forest

_____ (3 points)

7. If primary productivity of forests is positively correlated with temperature, which forest type has the highest primary productivity?

Lowest productivity?

_____ (3 points)

8. Which two forest types are most vulnerable to human disturbance and damage?

_____ (4 points)

9. Which forest type is closest to Taiwan's tree line?

_____ (3 points)

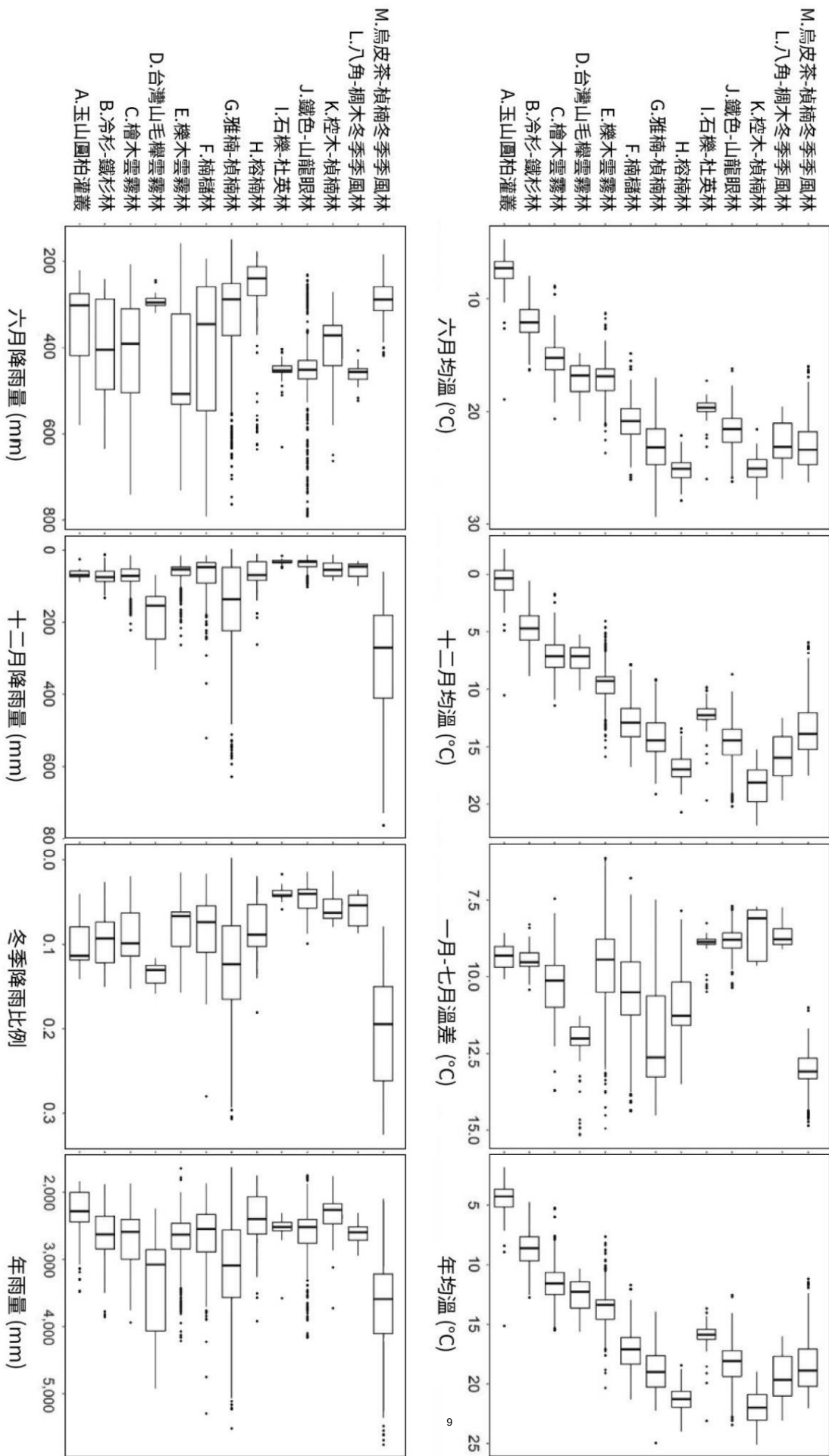
10. Among the three types of cloud forests, which forest type has the most limited distribution?

_____ (3 points)

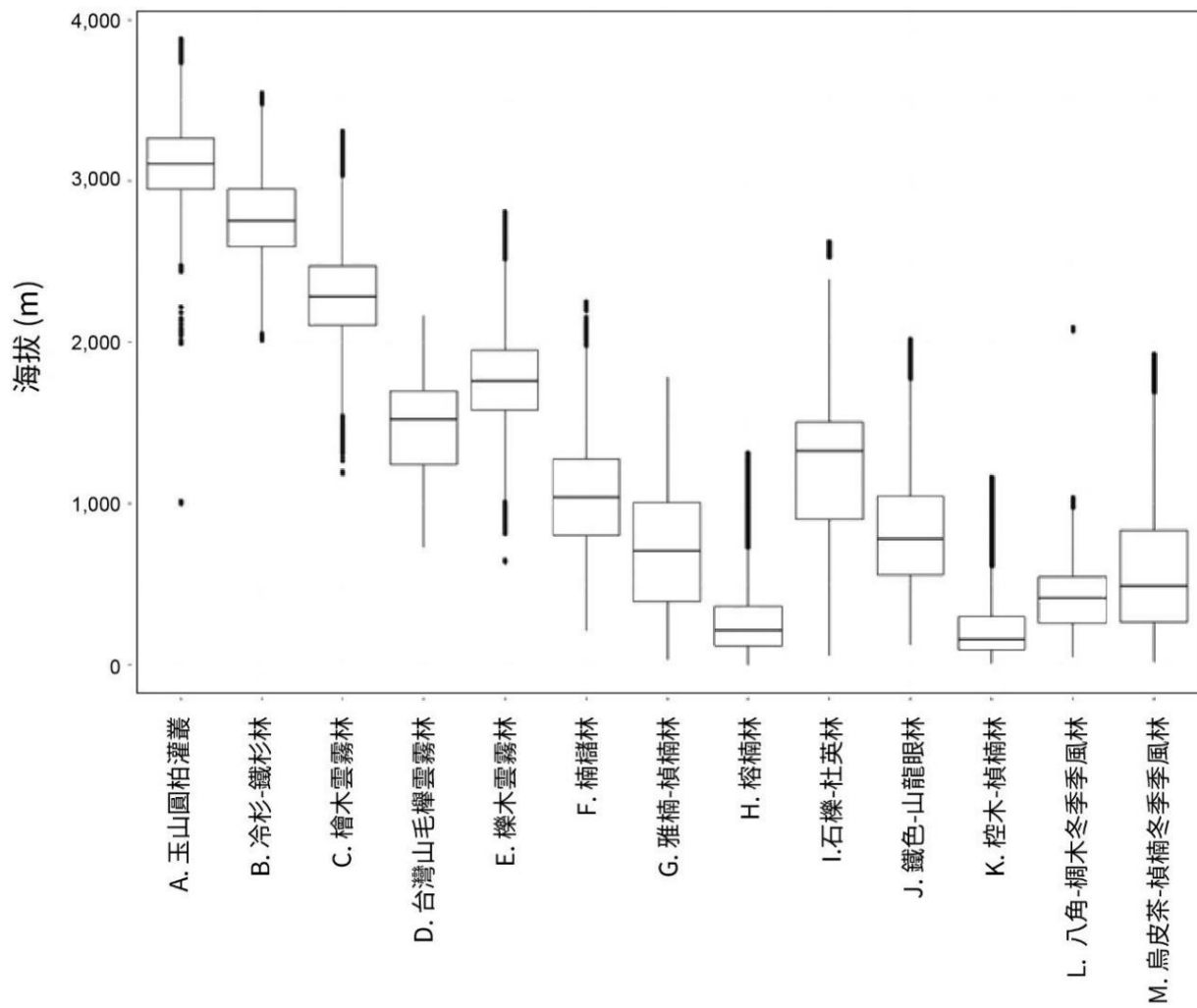
11. Forest types in subarctic regions located between 50 and 70 degrees north latitude (such as Siberia) are most similar to...

What are the two types of forests near Taiwan?

_____ (4 points)



(Figure 1) Climate and environmental conditions of the distribution of 13 plant community types in Taiwan.



(Figure 2) Altitude range of the sample areas for 13 plant community types in Taiwan